No.
Date

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Untuk dua buah persamaan : $f(x) = x^2 - x - 2 = 0$ dan $g(x) = x^2 - 9 = 0$ hitunglah pendekatan untuk mulai x sampai iterasi tiga metode resula falsi.

Jawab :

1. Persamaan $f(x) = x^2 - x - 2 = 0$

$f(1) = 1 - 1 - 2 = -2$; $a = 1$

$f(3) = 9 - 3 - 2 = 9$; $b = 3$

$$x_r = b - \frac{f(b)(b-a)}{f(b) - f(a)}$$

Iterasi 1 : $x_1 = 3 - \frac{9(3-1)}{9-(-2)} = 3 - \frac{8}{6} = 1,67$. $f(1,67) = -0,89$

Iterasi 2 : $x_2 = 3 - \frac{9(3-1,67)}{9-(-0,89)} = 3 - \frac{5,32}{9,89} = 1,91$. $f(1,91) = -0,27$

Iterasi 3 : $x_3 = 3 \frac{9(3-1,91)}{9-(-0,27)} = 3 - \frac{9,36}{9,27} = 1,98$

$x = 1,98$ // → hasil pendekatan

2. Persamaan $g(x) = x^2 - 9 = 0$

Interval awal

$g(1) = -3$; $a = 1$

$g(3)$ 5 ; $b = 3$

Iterasi 1 : $x_1 = 3 - \frac{5(3-1)}{5-(-3)} = 3 - \frac{10}{8} = 1,75$. $g(1,75) \approx -0,99$

Iterasi 2 : $x_2 = 3 - \frac{5(3-1,75)}{5-(-0,99)} = 3 - \frac{6,25}{5,99} \approx 1,95$. $g(1,95) \approx -0,20$.

Iterasi 3 : $x_3 = 3 - \frac{5(3-1,95)}{5-(-0,20)} = 3 - \frac{5,25}{5,20} \approx 1,99$

$x \approx 1,99$ → hasil pendekatan.

Kesimpulan :

$f(x) \rightarrow x \approx 1,98$

$g(x) \rightarrow x \approx 1,99$